Devansh Gupta

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Research Interests

I study statistical learning and optimization in settings where privacy, robustness, and memory constraints challenge standard methods, with a particular interest in the foundations of efficient and safe large-scale machine learning and generative AI, including large language models (LLMs).

Education

University of Southern California — **Los Angeles, California**
Ph.D. in Computer Science — *Aug 2023 – Expected May 2028*
Advised by Prof. Meisam Razaviyayn and Prof. Vatsal Sharan.
Relevant Coursework: Complexity Theory, Communication Complexity, Information Theory, Online Learning.

Indraprastha Institute of Information Technology, Delhi — **New Delhi, India**
B.Tech. in Computer Science and Artificial Intelligence — *Jul 2019 – May 2023*

Publications and Preprints

*These authors contributed equally.

- Ankit Pensia ⓓ **Devansh Gupta** ⓓ Kabir Aladin Verchand ⓓ Vatsal Sharan ⓓ Atreya Vedantam, "*Memory-Sample Tradeoffs in High-Dimensional Statistics*," Under submission.
- **Devansh Gupta**, Cristóbal Guzman, Enayat Ullah, "*Differentially Private Learning of Markov Chains: Near-Optimal Sample Complexity for Item and User Level Privacy*," Preprint, 2026. Link.
- Mahdi Salmani, Adel Javanmard, **Devansh Gupta**, Meisam Razaviyayn, "*Direct Pass@k Optimization for Thought Pruning: Balancing Quality and Diversity in Test-Time Reasoning*," Preprint, 2026. Link.
- Enayat Ullah, Sai Aparna Aketi*, **Devansh Gupta***, Huanyu Zhang, Meisam Razaviyayn, "*Efficient DP-SGD for LLMs with Randomized Clipping*," International Conference on Machine Learning (ICML '26); also presented at TPDP '26. Link.
- Alex Mulrooney*, **Devansh Gupta***, James Flemings, Huanyu Zhang, Murali Annavaram, Meisam Razaviyayn, Xinwei Zhang, "*Memory-Efficient Differentially Private Training with Gradient Random Projection*," Transactions on Machine Learning Research (TMLR '26). Link.
- **Devansh Gupta**, A. S. Poornash, Andrew Lowy, Meisam Razaviyayn, "*A Stochastic Optimization Framework for Private and Fair Learning From Decentralized Data*," Transactions on Machine Learning Research (TMLR '26); earlier version at PPAI@AAAI '25. Link.
- **Devansh Gupta**, Meisam Razaviyayn, Vatsal Sharan, "*On the Inherent Privacy of Zeroth-Order Projected Gradient Descent*," Artificial Intelligence and Statistics (AISTATS '25); **Oral** at OPT@NeurIPS '24. Link.
- Andrew Lowy, **Devansh Gupta**, Meisam Razaviyayn, "*Stochastic Differentially Private and Fair Learning*," International Conference on Learning Representations (ICLR '23); **Oral** at AFCP@NeurIPS '22. Link.
- **Devansh Gupta**, Aditya Saini, Sarthak Bhagat, Shagun Uppal, Rishi Raj Jain, Drishti Bhasin, Ponnurangam Kumaraguru, Rajiv Ratn Shah, "*A Suspect Identification Framework Using Contrastive Relevance Feedback*," Winter Conference on Applications of Computer Vision (WACV '23); related student abstract at AAAI '22. Link.
- **Devansh Gupta**, Vinayak Abrol, "*Time-Frequency and Geometric Analysis of Task-Dependent Learning in Raw Waveform-Based Acoustic Models*," 47th International Conference on Acoustics, Speech, and Signal Processing (ICASSP '22), 2022. Link.

Research Experience

Viterbi School of Engineering | University of Southern California — **Los Angeles, USA**
Graduate Research Assistant | Advisors: Meisam Razaviyayn, Vatsal Sharan — *Aug 2023 – Present*

Viterbi School of Engineering | University of Southern California — **Los Angeles, USA**
Visiting Research Scholar | Advisor: Andrew Lowy, Meisam Razaviyayn — *May 2022 – Jan 2023*

Cross-Caps Lab | IIIT Delhi — **New Delhi, India**
Undergraduate Researcher | Advisor: Vinayak Abrol — *Apr 2021 – Apr 2022*

Multimodal Digital Media Analysis Lab | IIIT Delhi — **New Delhi, India**
Research Intern | Advisor: Rajiv Ratn Shah — *Jan 2021 – Jan 2023*

Awards and Achievements

- One of 15 students selected across India for the IUSSTF-Viterbi Scholarship, providing full funding for a research internship at the Viterbi School of Engineering at USC.
- Dean's List for Academic Excellence for academic years 2019–2020, 2020–2021, 2021–2022, and 2022–2023.

Talks

- *On the Inherent Privacy of Two-Point Zeroth-Order Projected Gradient Descent*
  - USC Theory Lunch 2024
  - OPT@NeurIPS 2024
  - ICCOPT 2025

Teaching Experience

Teaching Assistant | CSCI 618: Trustworthy Machine Learning from an Optimization Lens, USC *Aug 2026 – Present*
Teaching Assistant | CSCI 570: Advanced Analysis and Design of Algorithms, USC *Aug 2025 – Dec 2025*
Teaching Assistant | Machine Learning (PG), IIIT Delhi *Aug 2022 – Dec 2022*
Teaching Assistant | Machine Learning (UG + PG), IIIT Delhi *Jan 2022 – May 2022*

Mentorship and Leadership Experience

Ph.D. Mentor | IUSSTF Viterbi-India Program, USC *May 2026 – Jul 2026*
Ph.D. Mentor | IUSSTF Viterbi-India Program, USC *May 2025 – Jul 2025*
Mentor | USC SHINE Program *May 2024 – Jun 2024*
Ph.D. Mentor | IUSSTF Viterbi-India Program, USC *May 2024 – Jul 2024*
Coordinator | Robotics Club, IIIT Delhi *Jan 2020 – Jun 2022*

Reviewing Experience

- ICLR 2026; NeurIPS 2023, 2024, and 2026 (external); ISIT 2024 (external).